

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology



Department of Mechatronics Engineering

Course No.: MTE 3104

Course Title: Microcontroller & Interfacing Sessional

Experiment No.: 2

Experiment Name: Implementation of a NAND Logic Gate Using ATmega32 Microcontroller

Remarks

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Experiment No.: 2

Experiment Name: Implementation of a NAND Logic Gate Using ATmega32 Microcontroller

Objectives:

- To study the basic architecture of the AVR ATmega32 microcontroller.
- To understand the function of I/O ports in the ATmega32.
- To write and execute a C program to control LEDs connected to PORTC using NAND gate at PORTD.
- To understand the logical operation of a NAND gate, to implement digital logic using a microcontroller, and to verify the NAND truth table using hardware simulation.

Apparatus Required:

- Proteus Simulation Software
- Atmel Studio

Introduction:

Digital logic gates are the fundamental building blocks of digital electronics and computer systems. Among all logic gates, the NAND gate holds special importance because it is a universal gate, meaning that any other logic gate or digital circuit can be constructed using only NAND gates [1]. Microcontrollers such as the ATmega32 provide a flexible platform for implementing digital logic functions through software, eliminating the need for discrete logic ICs and enabling rapid prototyping [2]. In this experiment, an ATmega32 microcontroller is used to implement a NAND gate by reading two digital inputs from push-button switches and producing the corresponding output on an LED. The logic operation is performed programmatically inside the microcontroller, demonstrating the use of embedded systems for digital logic realization.

Theory:

A NAND gate performs the logical NOT of the AND operation. The ATmega32 is an 8-bit AVR microcontroller featuring 32 programmable I/O pins, which can be configured as either inputs or outputs through software registers [2]. By reading the state of two input pins connected to switches and controlling an output pin connected to an LED, the NAND logic function can be implemented entirely in software. The output of the NAND gate is LOW only when both inputs are HIGH; otherwise, the output remains HIGH [1]. The truth table of a two-input NAND gate is shown below:

$$Y = A \cdot \bar{B}$$

Input A	Input B	Output Y
0	0	1
0	1	1
1	0	1
1	1	0

Fig 1: Truth table and logic of NAND gate

Code:

```
/*  
* GccApplication3.c  
* Created: 19/1/2026 8:07:37 am
```

```

* Author : Bayazid HS
*/
#include <avr/io.h>
int main(void)
{
    DDRC = 0b00000001; // PC0 as output
    DDRD = 0x00; // PORTD as input
    while (1)
    {
        if ( (PIND & (1<<PD0)) && !(PIND & (1<<PD1)) ) {
            PORTC = 0b00000001; // LED ON
        }
        else if ( !(PIND & (1<<PD0)) && (PIND & (1<<PD1)) ) {
            PORTC = 0b00000001; // LED ON
        }
        else if ( (PIND & (1<<PD0)) && (PIND & (1<<PD1)) ) {
            PORTC = 0b00000000; // LED OFF
        }
        else {
            PORTC = 0b00000001; // LED ON
        }
    }
    return 0;
}

```

Circuit Description:

The circuit consists of an ATmega32 microcontroller, two SPST switches used as logic inputs, two pull-down resistors to ensure stable logic levels, and an LED connected as the output indicator. The switches are connected to digital input pins of the microcontroller, while the LED is connected to a digital output pin through a current-limiting resistor. When the switches are pressed or released, the microcontroller reads the input states and computes the NAND operation, updating the LED state accordingly.

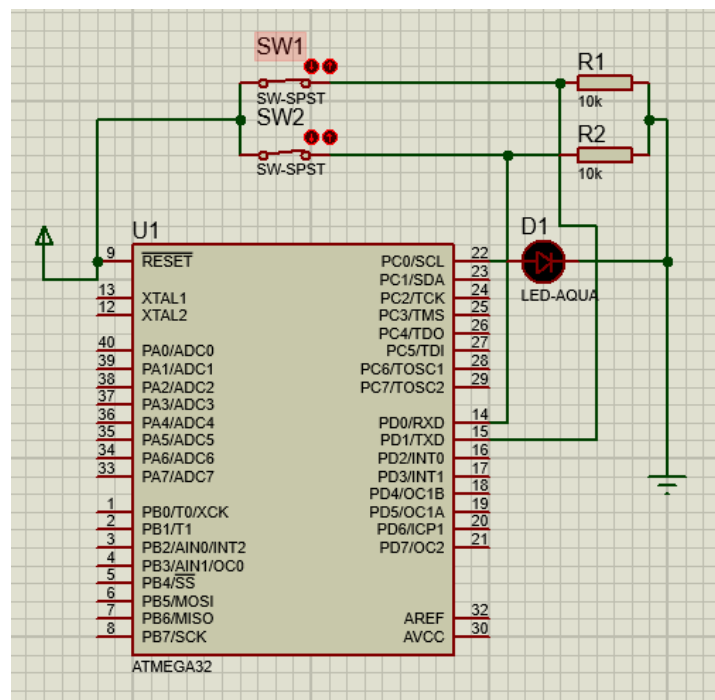


Fig 2: Circuit Diagram of NAND Gate in Proteus

Workings:

When the circuit is powered, the ATmega32 continuously monitors the logic levels of the two input pins connected to the switches. These inputs represent logic '0' when the switch is open and logic '1' when the switch is pressed. The microcontroller executes a NAND logic operation in software. If both inputs are HIGH, the output pin is driven LOW, turning the LED OFF. For all other input combinations, the output pin is driven HIGH, turning the LED ON. This behavior confirms the correct operation of the NAND gate.

Result:

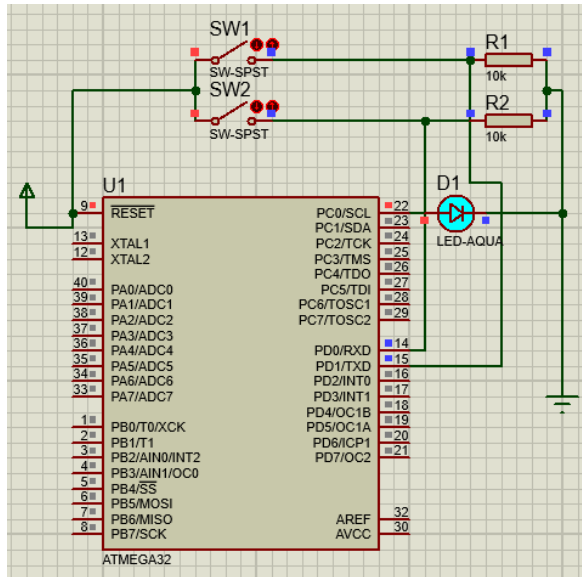


Fig 3: When two switches are OFF

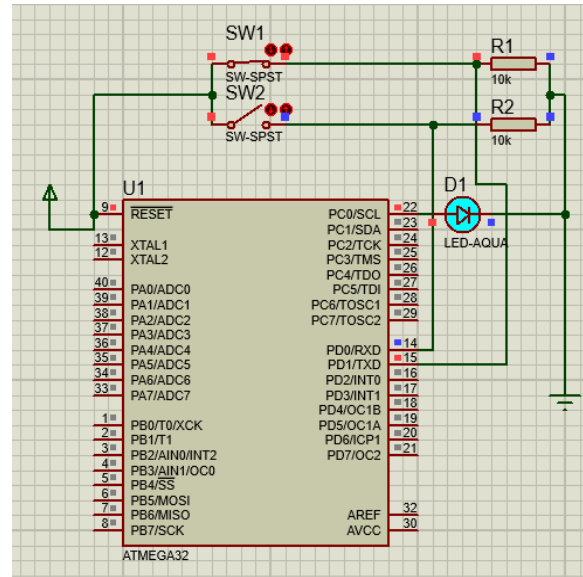


Fig 4: When switch 1 is ON

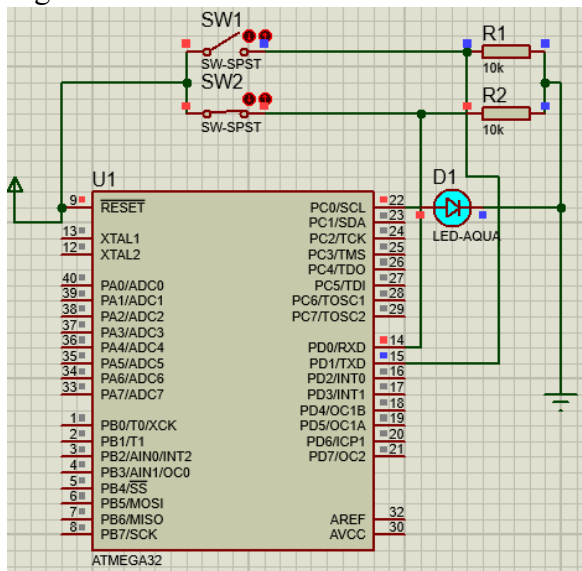


Fig 5: When switch 2 is ON

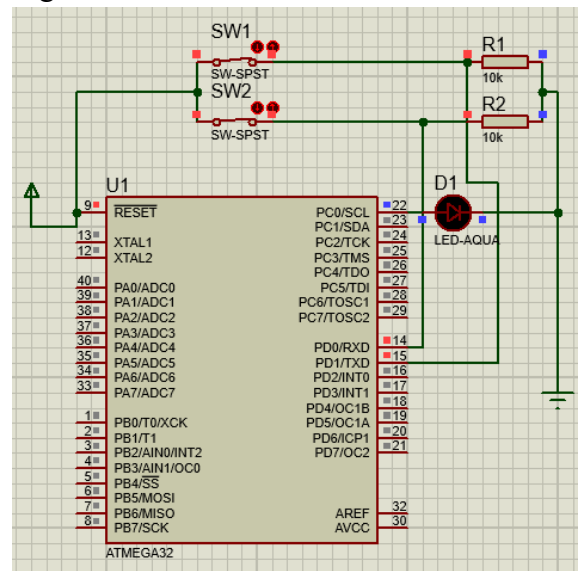


Fig 6: When two switches are ON

The experimental results confirm that the implemented circuit behaves as a NAND gate. The LED remains ON for all input combinations except when both switches are pressed simultaneously, at which point the LED turns OFF. This verifies the logical expression $Y = A \cdot B$.

Discussion:

The experiment demonstrates that microcontrollers can effectively replace traditional logic gate ICs in digital system design. Implementing logic functions through software provides greater flexibility, reduces hardware complexity, and allows easy modification of logic behaviour without changing physical components. However, compared to dedicated logic ICs, microcontroller-based logic may introduce slight processing delays and requires proper programming knowledge.

Conclusion:

In this experiment, a NAND gate was successfully implemented using an ATmega32 microcontroller, two switches, and an LED. The circuit operation was verified through Proteus simulation, and the observed results matched the theoretical NAND gate truth table. The experiment highlights the versatility of microcontrollers in digital logic applications and reinforces the fundamental concept of universal logic gates.

References

1. M. Morris Mano and M. D. Ciletti, *Digital Design*, 5th Edition, Pearson Education.
2. Atmel Corporation, *ATmega32 Datasheet*, Microchip Technology Inc.
3. Labcenter Electronics, *Proteus Design Suite User Guide*.